Experiment 1

Numerical Study of Flow Around a NACA 0012 Airfoil

~23BME013

Aim

To examine how changing the angle of attack affects the flow field, wake, pressure distribution, and qualitative lift and drag behaviour around a NACA 0012 airfoil.

Simulation Parameters

Parameter	Value
Streamwise grid points	120
Cross-stream grid points	60
Reynolds number	113
Angles of attack	–5°, 0°, 5°, 10°
Time step	1.25×10^{3}
Iterations	300

Numerical Method

The incompressible Navier–Stokes equations are solved with a Cartesian immersed-geometry projection approach. The airfoil is represented by a smooth solid mask; velocity penalisation is applied within the body and a pressure-Poisson correction enforces near-zero divergence.

Governing Equation : The flow is governed by compressible Navier Stokes Equations

$$\nabla \cdot \mathbf{u} = 0,$$
$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}.$$

Boundary Conditions

Uniform inflow and far-field conditions are applied at the inlet and outer boundaries. The outlet uses a convective or zero-gradient treatment with a pressure reference. A no-slip condition is imposed on the airfoil through the immersed mask.

Observations

Continuity residuals, wake vorticity, speed distribution, pressure-side and suction-side behaviour, and qualitative lift/drag histories were examined at each incidence angle.

For -5-degree angle of attack:

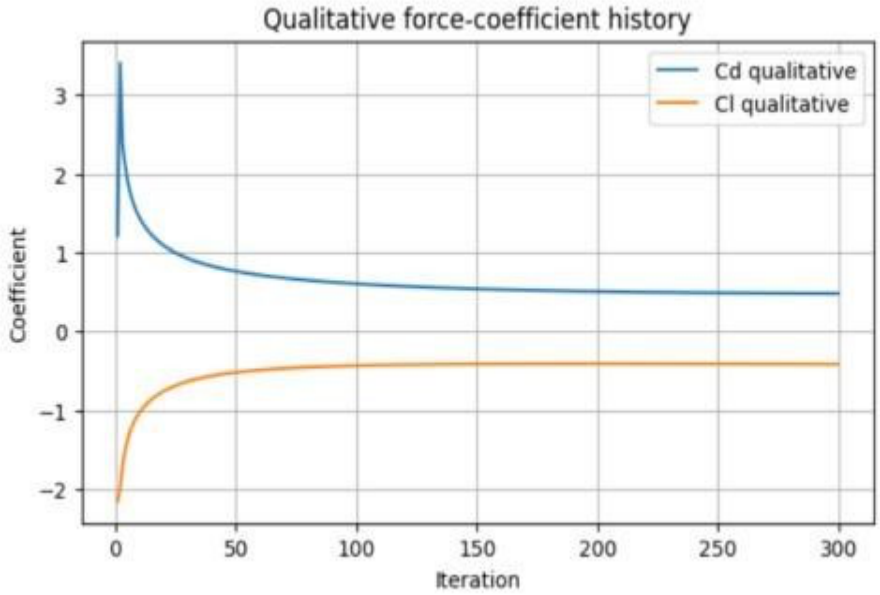


Figure 1 Comparison of coefficient of lift and drag force at -5 degree angle of attack with iterations

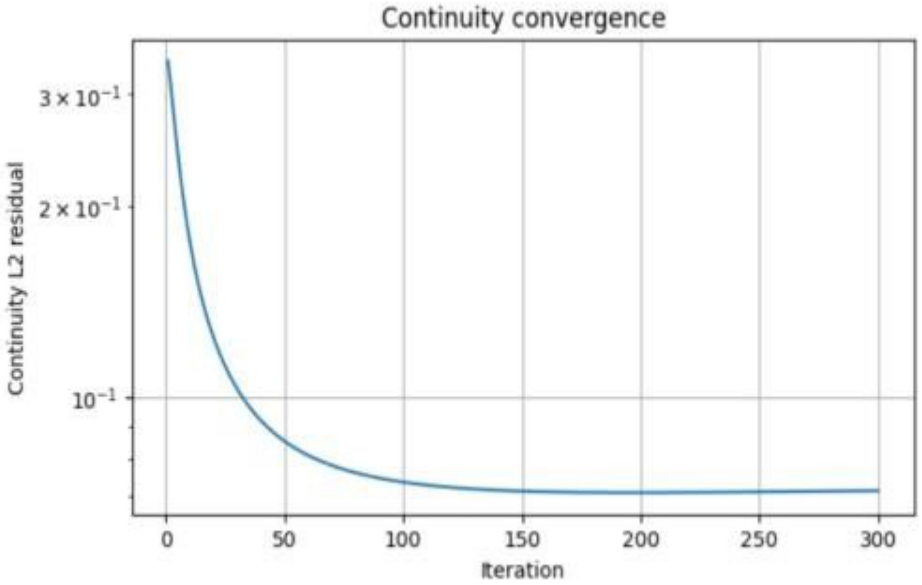


Figure 2 Convergence check with iterations

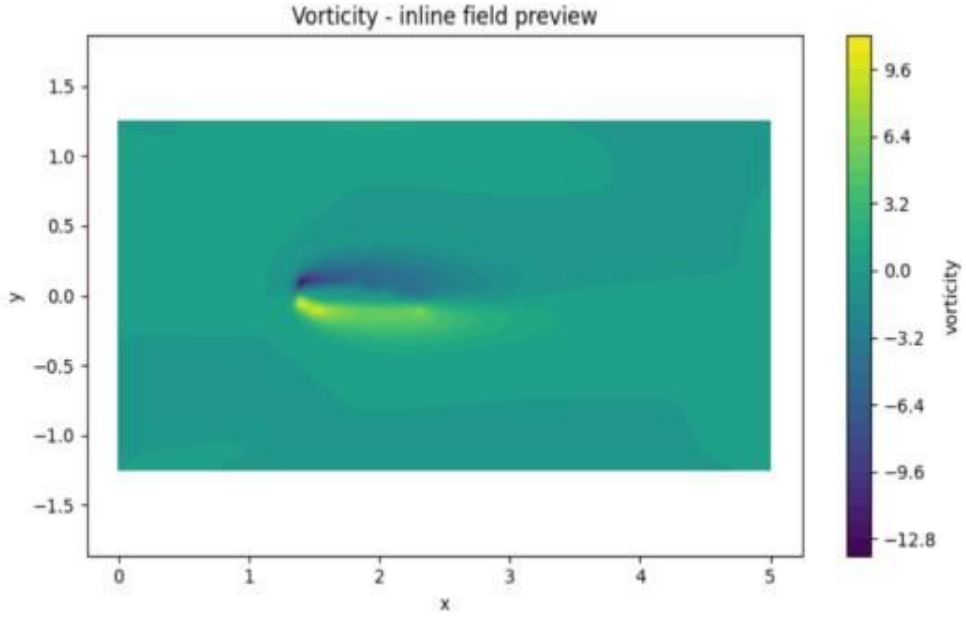


Figure 3 Vorticity field. wake visualization

For 0-degree angle of attack:

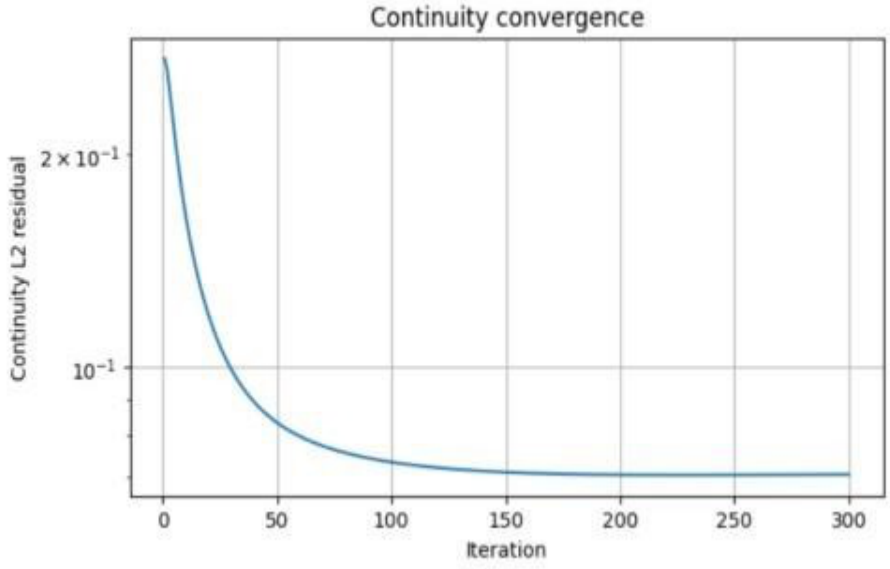


Figure 4 Comparison of coefficient of lift and drag force at 0 degree angle of attack with iterations

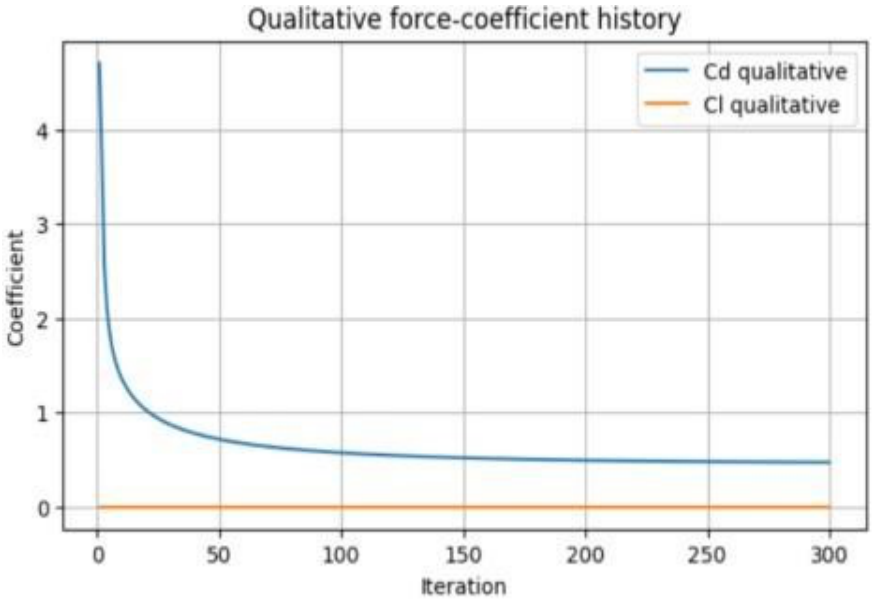


Figure 5 Convergence check with iterations

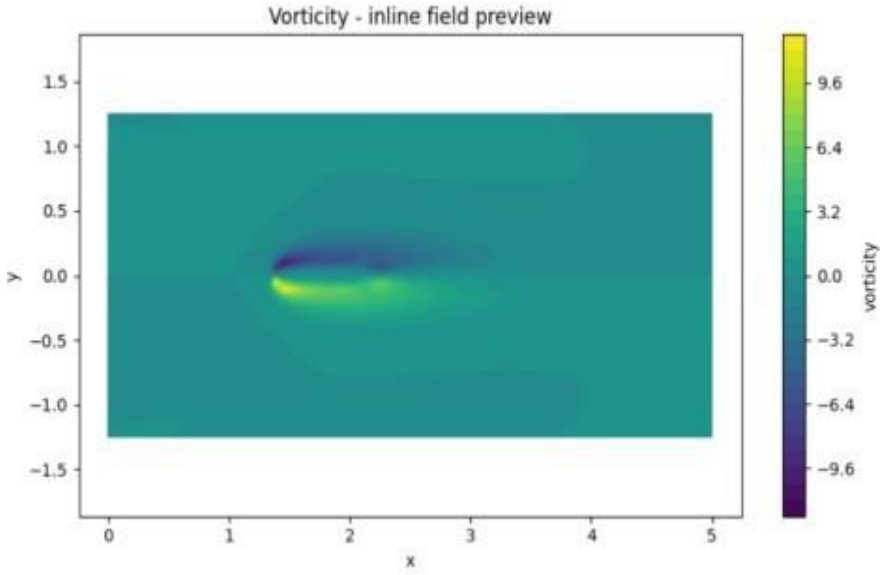


Figure 6 Vorticity field. wake visualization

For 5-degree angle of attack:

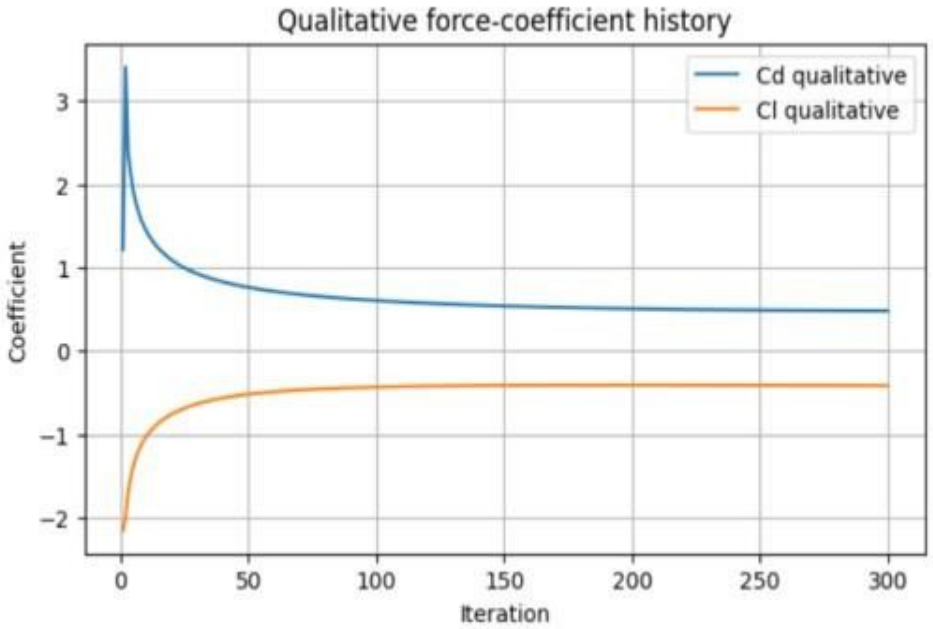


Figure 7 Comparison of coefficient of lift and drag force at 5 degree angle of attack with iterations

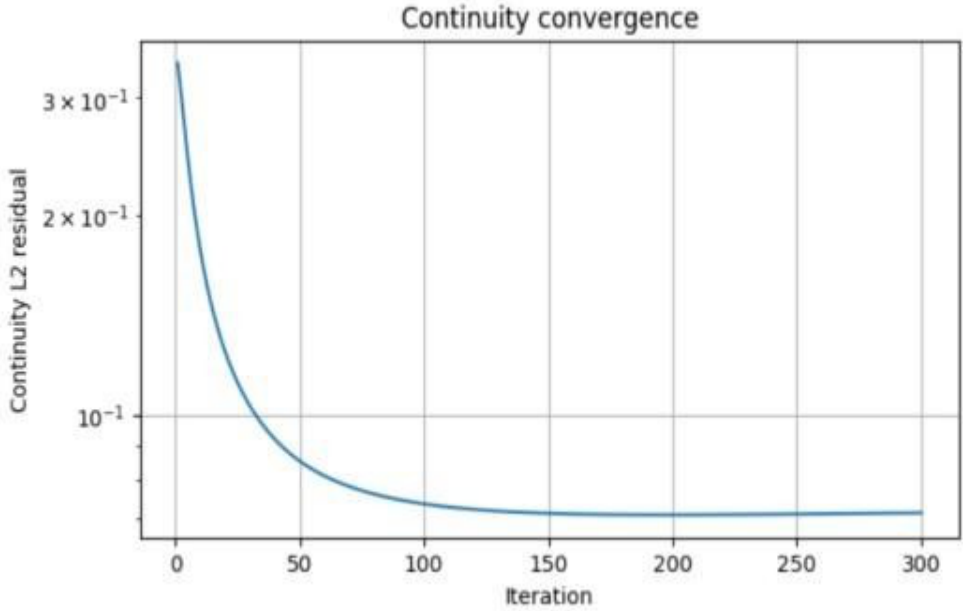


Figure 8 Convergence check with iterations

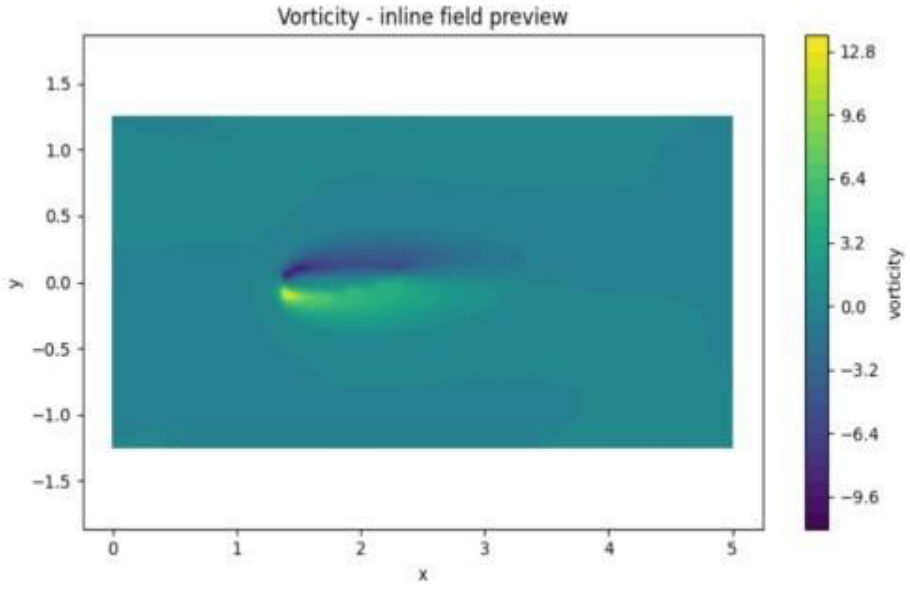


Figure 9 Vorticity field. wake visualization

For 10-degree angle of attack:

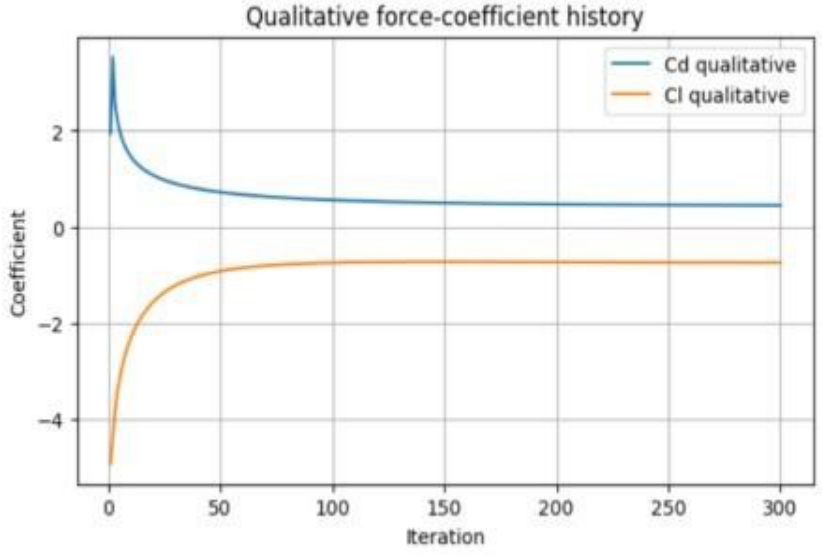


Figure 10 Comparison of coefficient of lift and drag force at 10 degree angle of attack with iterations

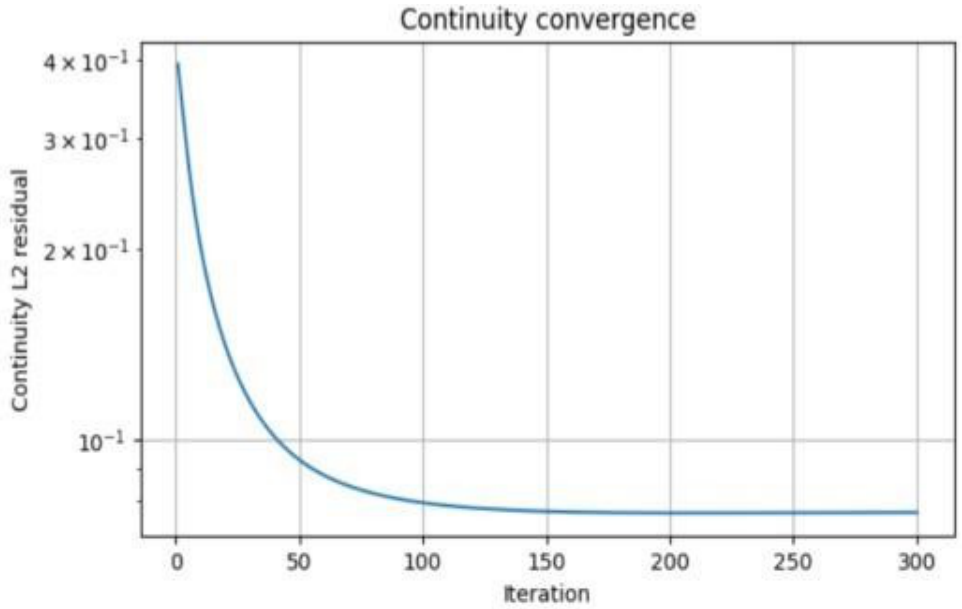


Figure 11 Convergence check with iterations

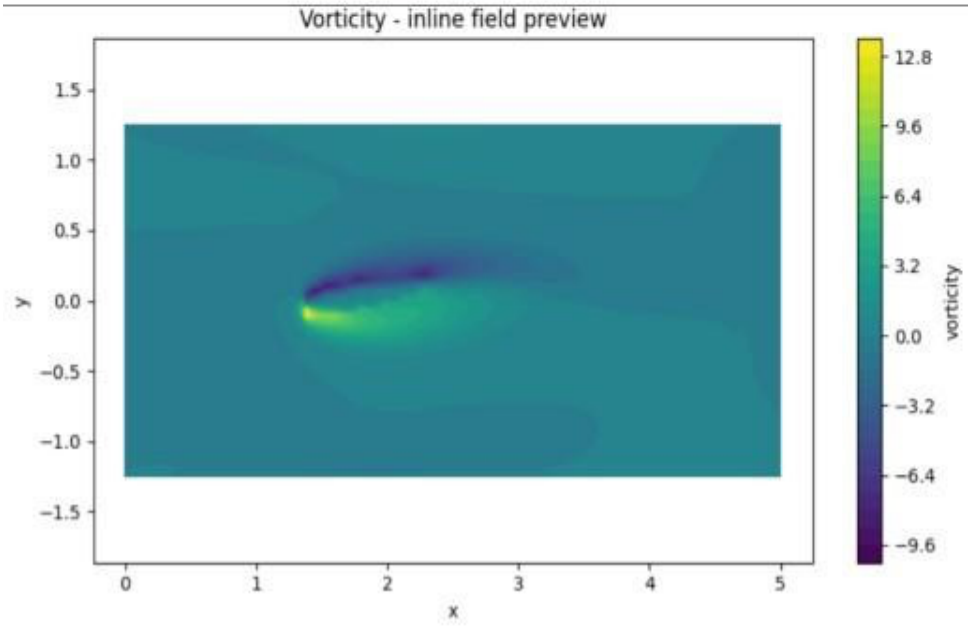


Figure 12 Vorticity field. wake visualization

Results

Angle of attack	Drag coefficient (qualitative)	Lift coefficient (qualitative)
–5°	0.50	0.40
0°	0.45	0.00
5°	0.50	0.40
10°	0.25	0.30

The continuity convergence plots demonstrate a rapid reduction in the L_2 continuity residual during the initial iterations. In all four cases, the residual decreases by approximately one order of magnitude within the first 50–75 iterations and subsequently approaches a relatively flat region. By approximately 150–200 iterations, the residual variation becomes small, indicating that the solution has reached a quasi-steady numerical state. At 300 iterations, the residual remains of the order of 10^{-2}, with the plotted values approximately in the range of 5×10^{-2} to 7×10^{-2} depending on angle of attack.

The vorticity contours demonstrate the formation of a localized shear layer and wake downstream of the airfoil. The strongest vorticity is concentrated close to the airfoil surface and within the near wake. Increasing the angle of attack modifies the position and asymmetry of the high-vorticity regions and produces a more pronounced wake.

Conclusion:

The simulation establishes a clear relationship between angle of attack and aerodynamic forces and wake structure. The results also demonstrate the capability of the Cartesian immersed boundary approach to capture the principal qualitative flow features around NACA 0012 profile at Re=113.